

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

<i>Criteria</i>	Understandin g of Data (2.5)	Analysis (2.5)	Applicatio n of Concepts (2)	Conclusion (1.5)	Clarity (1)	<i>Total(10)</i>
<i>Weightag e</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I M MEGHANADH(192325026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MEGHANADH

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

K-Means Clustering and Elbow Method

The **Elbow Method** is used to determine the optimal number of clusters by observing how the **SSE (Sum of Squared Errors)** decreases as the number of clusters increases.

Given:

K	SSE
1	600
2	350
3	220
4	200
5	190

1. Interpreting the Elbow Point

The SSE decreases significantly up to **K = 3**:

- K=1 → K=2: **600 → 350** (large decrease)
- K=2 → K=3: **350 → 220** (large decrease)
- K=3 → K=4: **220 → 200** (small decrease)
- K=4 → K=5: **200 → 190** (very small decrease)

Therefore, the **elbow point** is **K = 3**.

Optimal number of clusters = 3.

2. Patient Segmentation

Using **3 clusters**, patients can be grouped according to similarities in their **monthly medical expenses and health risk scores**.

Based on the given data, the clusters would generally represent groups such as:

- **High-risk / relatively lower expense patients** – patients requiring closer health monitoring.
- **Moderate-risk / moderate-expense patients** – patients requiring regular healthcare management.
- **Lower-risk / high-expense patients** – patients who may require analysis of why their medical expenses are high despite lower risk scores.

The exact cluster membership would be determined by running K-Means on the two features after appropriate scaling.

3. Effect on Healthcare Decision-Making

Selecting **K = 3** provides a useful balance between simplicity and meaningful segmentation.

- **Resource allocation:** Hospitals can prioritize staff, beds, and medical resources for high-risk groups.
- **Patient monitoring:** High-risk patients can receive more frequent check-ups and preventive care.
- **Cost management:** High-expense groups can be analyzed to identify reasons for increased medical spending.
- **Personalized healthcare:** Different treatment and follow-up strategies can be designed for different patient segments.
- **Avoids over-segmentation:** Choosing K=4 or K=5 provides only small SSE improvements, while creating additional groups that may not provide significant practical benefits.

Conclusion

The **optimal number of clusters is 3**, because the SSE shows a clear **elbow at K=3**.

Increasing the number of clusters beyond 3 produces only a small reduction in SSE. Thus, a **3-cluster solution** provides an effective and interpretable patient segmentation that can support healthcare resource allocation, risk monitoring, and cost-management decisions.

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Hierarchical Clustering

Hierarchical clustering groups similar data points into clusters by progressively merging the closest observations. A **dendrogram** represents these merging steps visually.

For the given dataset:

Order ID	Category	Order Value (₹)
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Order ID	Category	Order Value (₹)
O1	Electronics	55,000
O2	Clothing	25,000
O3	Electronics	57,000
O4	Furniture	30,000
O5	Clothing	27,000

1. Cluster Formation

The orders can be grouped based on similarities in **product category and order value**.

- **O1 and O3** are very similar because both are Electronics orders and their values are ₹55,000 and ₹57,000.
- **O2 and O5** are similar because both are Clothing orders and their values are ₹25,000 and ₹27,000.
- **O4** is a Furniture order with a value of ₹30,000, so it is less similar to the other orders.

The dendrogram would therefore first merge **O1 with O3** and **O2 with O5** because they have high similarity.

A suitable cut of the dendrogram gives **3 clusters**:

- **Cluster 1:** {O1, O3} → High-value Electronics orders
- **Cluster 2:** {O2, O5} → Medium-value Clothing orders
- **Cluster 3:** {O4} → Furniture order

2. Appropriate Number of Clusters

The appropriate number of clusters is **3** because it provides meaningful groups while preserving the major differences between product categories and order values.

If the dendrogram is cut at a suitable distance level, three distinct groups can be identified.

3. Significance for Customer Purchasing Behavior

These clusters help the retailer understand different purchasing patterns:

Cluster 1 – High-value Electronics:

Customers purchasing electronics tend to place **high-value orders**. The retailer can promote premium electronics, accessories, warranties, and related products.

Cluster 2 – Clothing:

Customers in this group make **moderate-value clothing purchases**. Discounts, seasonal offers, and product recommendations can encourage repeat purchases.

Cluster 3 – Furniture:

This represents a separate purchasing pattern involving furniture. The retailer can analyze furniture customers separately because their needs and purchasing behavior differ from clothing and electronics customers.

Conclusion

Hierarchical clustering identifies **3 meaningful clusters: high-value Electronics, Clothing, and Furniture**. The clusters allow the retailer to identify purchasing patterns, design targeted marketing strategies, recommend relevant products, and allocate resources more effectively.

Q3. Dataset: Mobile App User Behavior

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

HAC and Probability-Based Soft Clustering

The company wants to segment users based on **Daily Usage Time, Login Frequency, and In-App Purchase**. In probability-based clustering, a user does not have to belong completely to only one cluster. Instead, the user can have a **membership probability** for multiple clusters.

1. User Behavior Analysis

User Usage (min) Logins Purchase (₹) Behavior

U1	120	18	2,000	Highly active, moderate spending
U2	60	5	5,000	Low activity, high spending
U3	140	22	1,500	Highly active, lower spending
U4	50	4	6,000	Low activity, highest spending
U5	130	20	1,800	Highly active, moderate spending

The users naturally show two major behavior patterns:

- **Active users:** U1, U3, U5 have high usage time and frequent logins.
- **Low-activity/high-spending users:** U2 and U4 have lower usage and login frequency but higher purchases.

2. Probability-Based Cluster Membership

Suppose the HAC/soft-clustering process identifies two clusters:

- **Cluster A:** Highly Active Users
- **Cluster B:** High-Spending/Low-Activity Users

Illustrative membership probabilities could be:

User Cluster A: Active Cluster B: High-Spending

U1	0.90	0.10
U2	0.15	0.85
U3	0.95	0.05
U4	0.05	0.95
U5	0.92	0.08

For example, **U1 has 90% membership in Cluster A and 10% in Cluster B**, so U1 is mainly considered an active user. Similarly, **U2 has 85% membership in Cluster B**, indicating that the user behaves primarily like a high-spending, low-activity customer.

The probabilities are useful because users near the boundary between clusters can have significant membership in **both clusters**.

3. Advantages of Soft Clustering over Hard Clustering

Hard clustering:

Each user is assigned to exactly one cluster.

Soft clustering:

Each user receives a probability or membership value for each cluster.

Soft clustering is advantageous because:

1. **Captures mixed behavior:** A user can simultaneously show characteristics of different customer groups.
2. **Handles uncertain users:** Users near cluster boundaries are not forced into a single category.
3. **Better personalization:** Marketing strategies can consider multiple behavioral patterns.

4. **Improves recommendations:** Products, offers, and notifications can be selected according to the user's membership probabilities.
5. **Supports customer retention:** Users gradually changing their behavior can be detected more effectively.

Conclusion

Probability-based soft clustering provides a more flexible representation of mobile-user behavior than hard clustering. In this dataset, **U1, U3, and U5** mainly represent highly active users, while **U2 and U4** represent low-activity but high-spending users. Soft membership values help the company identify users with mixed or changing behavior and design more personalized engagement strategies.

Q4. Dataset: Agricultural Crop Monitoring

Farm Rainfall (mm) Soil Fertility Score Crop Yield (tons)

F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Effect of Normalization on Clustering

The agricultural research center uses clustering based on **Rainfall, Soil Fertility Score, and Crop Yield**. Since these variables have different numerical scales, normalization is important before applying a clustering algorithm such as K-Means.

1. Original Dataset

Farm	Rainfall (mm)	Soil Fertility	Crop Yield
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

The variables have different ranges:

- Rainfall: **280–520 mm**
- Soil Fertility: **3–9**
- Crop Yield: **65–90 tons**

Rainfall has a much larger numerical scale than soil fertility.

2. Clustering Before Normalization

Without normalization, distance-based clustering may give **more importance to rainfall** because its numerical values are much larger.

For example:

- F1 and F4 have relatively similar rainfall values and high soil fertility/yield.
- F2 and F5 have high rainfall, low fertility, and lower yield.
- F3 has medium rainfall, fertility, and yield.

However, the clustering algorithm may mainly consider the differences in rainfall and may not adequately consider soil fertility.

Thus, the clusters may be **biased toward rainfall differences**.

3. Clustering After Normalization

Normalization transforms the features into comparable scales, commonly between **0 and 1** or into standardized values with mean 0 and standard deviation 1.

After normalization:

- Rainfall, soil fertility, and crop yield contribute more equally to distance calculations.
- Soil fertility and crop yield receive appropriate importance.
- Farms with similar overall agricultural characteristics are grouped together.

A meaningful grouping could be:

- **Cluster 1:** F1 and F4 → lower rainfall, high fertility, high yield.
- **Cluster 2:** F2 and F5 → high rainfall, low fertility, lower yield.
- **Cluster 3:** F3 → medium rainfall, medium fertility, medium-high yield.

4. Influence of Normalization

Normalization improves clustering because:

1. **Prevents scale dominance:** Large-valued features such as rainfall do not overpower smaller-valued features such as soil fertility.
2. **Improves distance calculations:** All features contribute more fairly to Euclidean distance.
3. **Produces meaningful clusters:** Farms are grouped based on their overall characteristics rather than mainly one variable.
4. **Improves reliability:** The resulting clusters better represent actual agricultural conditions.
5. **Supports better decisions:** Researchers can identify farms with similar rainfall, soil quality, and productivity and recommend suitable farming strategies.

Conclusion

Normalization is important when clustering agricultural data with features measured on different scales. **Before normalization, rainfall may dominate the clustering process. After normalization, rainfall, soil fertility, and crop yield have a more balanced influence.** Therefore, normalized clustering produces more reliable and meaningful farm segments, supporting better crop planning, resource allocation, and agricultural decision-making.

Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score

E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48
K=3 0.68
K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Silhouette Score Analysis

The company uses clustering to group employees based on **Productivity Score, Attendance, and Skill Assessment Score**. The **Silhouette Score** measures how well each employee fits within its assigned cluster compared with other clusters.

1. Given Silhouette Scores

Number of Clusters (K)	Silhouette Score
K = 2	0.48
K = 3	0.68
K = 4	0.52

A **higher Silhouette Score indicates better-defined and more separated clusters**. The score generally ranges from **-1 to +1**:

- Close to **+1** → well-separated clusters
- Around **0** → overlapping clusters
- Negative → employees may be assigned to inappropriate clusters

2. Comparison of Models

K = 2 → Score = 0.48

The clustering is moderately acceptable, but the employee groups may not be sufficiently separated. Some differences in productivity, attendance, and skills may be lost by using only two groups.

K = 3 → Score = 0.68

This gives the **highest Silhouette Score**. It indicates that the employees are relatively well grouped within their clusters and the clusters are clearly separated.

K = 4 → Score = 0.52

Although four clusters provide more detailed segmentation, the lower score indicates that some clusters may overlap or contain employees with similar characteristics.

3. Most Suitable Number of Clusters

Therefore:

Optimal number of clusters = K = 3

because:

0.68 > 0.52 > 0.48

The K=3 model provides the best balance between **cluster cohesion and separation**.

4. Possible Employee Segmentation

Using K=3, employees could broadly be interpreted as:

- **Cluster 1 – High Performers:** High productivity, attendance, and skill scores.
Example: E1 and E5.
- **Cluster 2 – Moderate Performers:** Moderate productivity, attendance, and skill levels.
Example: E3.
- **Cluster 3 – Employees Requiring Improvement:** Lower productivity, attendance, and skill scores.
Example: E2 and E4.

The exact cluster assignment would depend on the clustering algorithm and distance calculations.

5. Contribution to Employee Performance Evaluation

The selected **K=3 model** can help the company:

1. **Identify high performers** for recognition and rewards.
2. **Identify employees needing support** and provide appropriate training.
3. **Analyze attendance and productivity patterns** together.
4. **Design personalized development programs** based on skill gaps.
5. **Improve workforce planning** by understanding different employee groups.

Conclusion

The **K=3 clustering model is the most suitable** because it has the highest Silhouette Score of **0.68**, indicating well-separated and cohesive employee clusters. This segmentation helps management distinguish high-performing, average, and improvement-needed employees, supporting **training, recognition, workforce planning, and performance improvement**.